



Qualification Pack

PCB NPI Fabrication and Verification Specialist

QP Code: CUTM/ELE/Q0101

Version: 1.0

NSQF Level: 5

Centurion University of Technology and Management || Aluri Nagar, Paralakhemundi, R Sitapur Uppalada || email:partha@cutm.ac.in



Qualification Pack

Contents

CUTM/ELE/Q0101: PCB NPI Fabrication and Verification Specialist	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
CUTM/ELE/N0101: Interpretation of designs and drawings and process planning	5
CUTM/ELE/N0102: Copper CAM Programming, Metrology, Maintenance, and repair of machine.	10
CUTM/ELE/N0103: Operate PCB prototype machine and complete the machining process	17
CUTM/ELE/N0104: Inspect, Finalize, and deliver the parts.	24
CUTM/ELE/N0105: Implement organizational health, safety and security for self and others	29
DGT/VSQ/N0102: Employability Skills (60 Hours)	34
Assessment Guidelines and Weightage	41
<i>Assessment Guidelines</i>	41
<i>Assessment Weightage</i>	42
Acronyms	44
Glossary	45

Qualification Pack

CUTM/ELE/Q0101: PCB NPI Fabrication and Verification Specialist

Brief Job Description

PCB New Product Implementation (NPI) Fabrication and Verification Specialist responsible for understanding the design requirements, material, generate design program for PCB manufacturing of various electronic circuit and inspect parts to finalize and deliver the work piece. She/he is also responsible for regular maintenance of machines.

Personal Attributes

A PCB NPI Fabrication and Verification Specialist should be detail-oriented, analytical, and quality focused. They must have strong problem-solving skills, technical aptitude, and the ability to interpret complex designs. Precision, teamwork, adaptability to new technologies, and a commitment to meeting deadlines are essential traits for this advanced electronics role.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [CUTM/ELE/N0101: Interpretation of designs and drawings and process planning](#)
2. [CUTM/ELE/N0102: Copper CAM Programming, Metrology, Maintenance, and repair of machine.](#)
3. [CUTM/ELE/N0103: Operate PCB prototype machine and complete the machining process](#)
4. [CUTM/ELE/N0104: Inspect, Finalize, and deliver the parts.](#)
5. [CUTM/ELE/N0105: Implement organizational health, safety and security for self and others](#)
6. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Electronics
Sub-Sector	New product Integration
Occupation	PCB Fabrication and Verification
Country	India
NSQF Level	5

Qualification Pack

Credits	18
Aligned to NCO/ISCO/ISIC Code	NSQF/NCrF
Minimum Educational Qualification & Experience	B.E./B.Tech (Completed 3rd year in Electronics, Electrical, Mechanical, Production Engg.) with NA of experience OR B.E./B.Tech (pursuing 3rd year with continuing education in Electronics, Electrical, Mechanical, production) with NA of experience OR Completed 2nd year diploma after 12th (in Electronics, Electrical, Mechanical or production) with NA of experience OR Completed 3 year diploma after 10th (in Electronics, Electrical, Mechanical or production) with 1 Year of experience relevant experience OR Previous relevant Qualification of NSQF Level (4) with 3 Years of experience relevant experience OR Previous relevant Qualification of NSQF Level (4.5) with 1.5 years of experience relevant experience
Minimum Level of Education for Training in School	12th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	19 Years
Last Reviewed On	NA
Next Review Date	31/10/2026
NSQC Approval Date	01/11/2023
Version	1.0
Reference code on NQR	QG-05-AG-01265-2023-V1-CUTM
NQR Version	1.0

Qualification Pack

CUTM/ELE/N0101: Interpretation of designs and drawings and process planning

Description

Fabrication and Verification Specialist focuses on analyzing PCB schematics, Gerber files, and CAD drawings. It includes process planning, material selection, tool path optimization, and ensuring manufacturability while adhering to industry standards and quality requirements.

Scope

The scope covers the following :

- Fabrication and Verification Specialist focuses on analyzing PCB schematics, Gerber files, and CAD drawings. It includes process planning, material selection, tool path optimization, and ensuring manufacturability while adhering to industry standards and quality requirements.

Elements and Performance Criteria

Interpretation of designs and drawings

To be competent, the user/individual on the job must be able to:

- PC1.** interpret technical drawings and specifications of product.
- PC2.** identify critical features of product, connections, and geometric requirements.
- PC3.** analyze manufacturability with available resources.
- PC4.** identify and prepare for mental representation of parts.
- PC5.** determine materials for PCB prototype.
- PC6.** ascertain stock to be kept for different operations.
- PC7.** identify critical sequences with different tools.
- PC8.** plan the production of parts according to specifications.

Process Planning

To be competent, the user/individual on the job must be able to:

- PC9.** ascertain solutions with reference to capacities of working environment and required work (size of batch, complexity)
- PC10.** determine features of PCB board and appropriate machining and measuring processes.
- PC11.** validate and prepare best work holding methods.
- PC12.** identify, prepare, and calibrate correct measuring tools
- PC13.** identify and prepare correct cutting tools.
- PC14.** identify critical sections (with high possibility of damage or unsafe practice) and consider alternatives.
- PC15.** devise innovative ways of using environment to solve technical issues.
- PC16.** ensure if potential solutions will be reliable until end of process.
- PC17.** weigh each solution and choose the best (considering speed, safety, and price)
- PC18.** make a final choice and lock strategy.

Qualification Pack

- PC19.** plan operations and sequences (machining strategy) based on specified data.
PC20. create awareness actions for critical operations where no alternative is available.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** technical drawings, blueprints, symbols, and GD&T (Geometric Dimensioning & Tolerancing).
KU2. How to analyze 2D and 3D CAD designs, including formats like DWG, DXF, and Gerber.
KU3. material properties, tolerances, and selection criteria for manufacturing.
KU4. step-by-step workflow for converting designs into finished products.
KU5. suitable machinery, tools, and fixtures for different fabrication processes.
KU6. design inconsistencies and recommending improvements for manufacturability.
KU7. calculating material, labor, and production costs based on design requirements.
KU8. How to maintain process documentation, work instructions, and revision histories.
KU9. compliance with workplace safety norms and eco-friendly manufacturing practices.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate fabrication issues, test results, and quality concerns with teams and clients.
GS2. Identify PCB defects, troubleshooting issues, and implementing corrective actions.
GS3. Prioritize tasks in the PCB fabrication and verification process to meet production deadlines.
GS4. work effectively with design engineers, quality inspectors, and production teams.
GS5. do measurements, tolerances, and calculations related to PCB design and verification.
GS6. handle drilling, etching, lamination, and solder mask application.
GS7. prepare detailed reports on fabrication defects, quality issues, and test outcomes.
GS8. follow safety protocols, handling hazardous chemicals, and implementing electrostatic discharge (ESD) protection.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Interpretation of designs and drawings</i>	8	16	-	9
PC1. interpret technical drawings and specifications of product.	1	2	-	1
PC2. identify critical features of product, connections, and geometric requirements.	1	2	-	1
PC3. analyze manufacturability with available resources.	1	2	-	1
PC4. identify and prepare for mental representation of parts.	1	2	-	1
PC5. determine materials for PCB prototype.	1	2	-	1
PC6. ascertain stock to be kept for different operations.	1	2	-	2
PC7. identify critical sequences with different tools.	1	2	-	1
PC8. plan the production of parts according to specifications.	1	2	-	1
<i>Process Planning</i>	22	24	-	21
PC9. ascertain solutions with reference to capacities of working environment and required work (size of batch, complexity)	1	2	-	1
PC10. determine features of PCB board and appropriate machining and measuring processes.	1	2	-	2
PC11. validate and prepare best work holding methods.	2	2	-	2
PC12. identify, prepare, and calibrate correct measuring tools	2	2	-	2
PC13. identify and prepare correct cutting tools.	2	2	-	2
PC14. identify critical sections (with high possibility of damage or unsafe practice) and consider alternatives.	2	2	-	2

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. devise innovative ways of using environment to solve technical issues.	2	2	-	2
PC16. ensure if potential solutions will be reliable until end of process.	2	2	-	3
PC17. weigh each solution and choose the best (considering speed, safety, and price)	2	3	-	3
PC18. make a final choice and lock strategy.	2	2	-	-
PC19. plan operations and sequences (machining strategy) based on specified data.	2	2	-	1
PC20. create awareness actions for critical operations where no alternative is available.	2	1	-	1
NOS Total	30	40	-	30



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/ELE/N0101
NOS Name	Interpretation of designs and drawings and process planning
Sector	Electronics
Sub-Sector	
Occupation	PCB Fabrication and Verification
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	01/11/2023
Next Review Date	31/10/2026
NSQC Clearance Date	01/11/2023

Qualification Pack

CUTM/ELE/N0102: Copper CAM Programming, Metrology, Maintenance, and repair of machine.

Description

Fabrication and Verification Specialist covers programming PCB machining using Copper CAM, precision metrology for quality checks, preventive maintenance, and troubleshooting machine issues. It ensures efficient PCB fabrication, adherence to specifications, and optimal machine performance.

Scope

The scope covers the following :

- Fabrication and Verification Specialist covers programming PCB machining using Copper CAM, conducting precision metrology for quality checks, performing preventive maintenance, and troubleshooting machine issues. It ensures accurate PCB fabrication, optimal machine performance, and adherence to industry standards.

Elements and Performance Criteria

Create programming

To be competent, the user/individual on the job must be able to:

- PC1.** understand design requirements, technical specification, and application.
- PC2.** select best sequence for production of parts based on type and part specification.
- PC3.** use skill to specific software and related hardware.
- PC4.** program for managing isolation engraving, drilling, and cutting printed-circuit board prototypes.
- PC5.** generate programs using copper CAM software.
- PC6.** transfer programs to machines.
- PC7.** create or edit programs directly on the control.

Study metrology requirements

To be competent, the user/individual on the job must be able to:

- PC8.** prepare objects and measuring instruments for measurements.
- PC9.** calibrate, adjust, and align measuring instruments.
- PC10.** select correct measuring instruments and devices (styluses, probes, etc.), auxiliary and fixing devices (vices, V blocks, clamps, etc.), relating to measurement strategy
- PC11.** perform measurements using various control and measuring instruments.
- PC12.** read the indications of measuring instruments.
- PC13.** identify and estimate the correctness of measurements and the reliability of the data obtained, to minimize associated human factor errors.
- PC14.** find the required information in specialized reference books, tables, or diagrams.
- PC15.** carry out routine maintenance of measuring instruments.

Preparation and PCB machining

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC16.** choose the technology that best suits task.
- PC17.** choose material that best suits task.
- PC18.** prepare models for forming according to selected technology and material (placement, orientation, supports, shrinkage)
- PC19.** simulate and analyze forming processes.
- PC20.** start and control the build process.
- PC21.** assign necessary post-processing processes and define their complexity.

Maintenance and repair of machining equipment

To be competent, the user/individual on the job must be able to:

- PC22.** do preventative maintenance to avoid issues arising.
- PC23.** make repairs when problem arise.

Complete the documentation

To be competent, the user/individual on the job must be able to:

- PC24.** note all production related information on respective register, log, work sheet.
- PC25.** update all production information on MIS system.
- PC26.** store Copper CAM program against batch number for future reference.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** copper CAM tools for PCB design, milling, and engraving operations.
- KU2.** to interpret Gerber files.
- KU3.** efficient milling and drilling paths to minimize material wastage and improve accuracy.
- KU4.** calipers, micrometers, coordinate measuring machines (CMM), and optical inspection tools.
- KU5.** PCB dimensions, hole placements, and trace widths meet design specifications.
- KU6.** Regular lubrication, calibration, and part inspections to ensure optimal performance.
- KU7.** diagnosing and fixing errors in drilling, milling, and etching machines.
- KU8.** spindles, motors, controllers, and pneumatic/hydraulic systems in PCB manufacturing.
- KU9.** safety protocols to prevent electrical hazards, mechanical failures, and workplace accidents.
- KU10.** proper disposal of metal shavings, chemical waste, and adherence to green manufacturing practices.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** interpret and modify Gerber, NC Drill, and CAM files for PCB fabrication.
- GS2.** PCB milling and engraving process optimization in Copper CAM software.
- GS3.** do precision measurement techniques using CMM, calipers, micrometers.
- GS4.** Troubleshoot for diagnosing and resolving issues in PCB manufacturing machines.
- GS5.** do preventive maintenance practices for CNC and PCB fabrication machines.



Qualification Pack

- GS6.** report machine faults, quality issues, and process deviations.
- GS7.** do time management and task prioritization in PCB prototyping, inspection, and machine upkeep.
- GS8.** to follow safety protocols and ESD (Electrostatic Discharge) protection measures.
- GS9.** keep record for machine maintenance, repairs, and calibration.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Create programming</i>	12	22	-	-
PC1. understand design requirements, technical specification, and application.	1	2	-	-
PC2. select best sequence for production of parts based on type and part specification.	1	2	-	-
PC3. use skill to specific software and related hardware.	1	2	-	-
PC4. program for managing isolation engraving, drilling, and cutting printed-circuit board prototypes.	1	2	-	-
PC5. generate programs using copper CAM software.	6	10	-	-
PC6. transfer programs to machines.	1	2	-	-
PC7. create or edit programs directly on the control.	1	2	-	-
<i>Study metrology requirements</i>	14	16	-	-
PC8. prepare objects and measuring instruments for measurements.	2	2	-	-
PC9. calibrate, adjust, and align measuring instruments.	2	2	-	-
PC10. select correct measuring instruments and devices (styluses, probes, etc.), auxiliary and fixing devices (vices, V blocks, clamps, etc.), relating to measurement strategy	2	2	-	-
PC11. perform measurements using various control and measuring instruments.	1	2	-	-
PC12. read the indications of measuring instruments.	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. identify and estimate the correctness of measurements and the reliability of the data obtained, to minimize associated human factor errors.	2	2	-	-
PC14. find the required information in specialized reference books, tables, or diagrams.	2	2	-	-
PC15. carry out routine maintenance of measuring instruments.	2	2	-	-
<i>Preparation and PCB machining</i>	9	12	-	-
PC16. choose the technology that best suits task.	1	2	-	-
PC17. choose material that best suits task.	2	2	-	-
PC18. prepare models for forming according to selected technology and material (placement, orientation, supports, shrinkage)	2	2	-	-
PC19. simulate and analyze forming processes.	1	2	-	-
PC20. start and control the build process.	1	2	-	-
PC21. assign necessary post-processing processes and define their complexity.	2	2	-	-
<i>Maintenance and repair of machining equipment</i>	2	4	-	-
PC22. do preventative maintenance to avoid issues arising.	1	2	-	-
PC23. make repairs when problem arise.	1	2	-	-
<i>Complete the documentation</i>	3	6	-	-
PC24. note all production related information on respective register, log, work sheet.	1	2	-	-
PC25. update all production information on MIS system.	1	2	-	-
PC26. store Copper CAM program against batch number for future reference.	1	2	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	40	60	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/ELE/N0102
NOS Name	Copper CAM Programming, Metrology, Maintenance, and repair of machine.
Sector	Electronics
Sub-Sector	
Occupation	PCB Fabrication and Verification
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	01/11/2023
Next Review Date	31/10/2026
NSQC Clearance Date	01/11/2023

Qualification Pack

CUTM/ELE/N0103: Operate PCB prototype machine and complete the machining process

Description

Fabrication and Verification Specialist focuses on setting up, calibrating, and operating PCB prototype machines. It includes tool path optimization, precision machining, quality inspection, troubleshooting, and ensuring compliance with safety and industry standards for efficient PCB fabrication.

Scope

The scope covers the following :

- Fabrication and Verification Specialist focuses on setting up, calibrating, and operating PCB prototype machines. It includes tool path optimization, precision machining, quality inspection, troubleshooting, and ensuring compliance with safety and industry standards for efficient PCB fabrication.

Elements and Performance Criteria

Operate PCB prototype machining

To be competent, the user/individual on the job must be able to:

- PC1.** follow selected process strategy.
- PC2.**
 - follow a given process strategy (external CAM programme)
 - opening Gerber file containing one circuit layer
 - opening next layers (maximum 4), if any
 - detecting or plotting card cut-out contour
 - opening Excellon file for drillings, if any
 - aligning layers (automatic or manual)
 - plotting tracks that are centerline texts
 - calculating isolation contours
- PC3.** upload generated CAM programme to PCB prototype machine and perform test runs.
- PC4.** load G-Code for Top Layer PCB prototype machine and print the circuit.
- PC5.** stick two-layer FR4 Material (Copper board) on Machine bed using 2way tape or any other clamp.
- PC6.** mount and align selected bare PCB board.
- PC7.** plug engraving bit in spindle motor.
- PC8.** connect one terminal of leveler probe in engraving bit and another one terminal of leveler probe stick on FR4 material.
- PC9.** using the Hand-Remote adjusts the X, Y and Z axis and set origin point.
- PC10.** select X-axis and press “=0” button and set X = 0, Select Y-axis and press “=0” button and set Y = 0, select Z-axis and rotate hand-wheel in Hand-Remote and move to 2mm. Now the Z = 2
- PC11.** press the mode button and select PC mode.
- PC12.** press the “cycle start” button in PCB prototype machine and start the auto surface detection operation.

Qualification Pack

- PC13.** completing the surface detection, it will come to starting position with Z = 20mm. Then remove the probe and press “cycle start” button and it will start Engraving. After completing the Top Layer it will come to starting position.
- PC14.** change the drill bit and drill four corner of your circuit board with bed.
- PC15.** remove the copper board from bed and flip the copper. Fit the copper board on the bed with correct position.
- PC16.** set the origin position. Select the mode button to Hand Remote mode. Select “X” move to 80mm and press “=0” button
- PC17.** load G-Code for Bottom Layer and print your circuit.
- PC18.** plug engraving bit in spindle motor. And connect one terminal of leveler probe in engraving bit and another one terminal of leveler probe stick on FR4 material.
- PC19.** press “Auto detect” button in hand-Remote and the Z axis move and touch on surface of the copper glad and came up 10mm top from the surface.
- PC20.** select Z-axis and rotate hand-wheel in Hand-Remote and move to 2mm. Now the Z = 2
- PC21.** press mode button and select PC mode.
- PC22.** press the “cycle start” button in Mach3mill and start the auto surface detection operation.
- PC23.** after completing the surface detection, it will come to starting position with Z = 20mm. Then remove probe and press “cycle start” button and it will start Engraving.
- PC24.** after completing the Bottom Layer it will come to starting position.
- PC25.** load G-Code for Drilling & repeat the above process from PC 18 to 24.
- PC26.** load G-Code for Cutting & repeat the above process from PC 18 to 24
- PC27.** remove the PCB board and separate machined PCB circuit board.

Complete the machining process

To be competent, the user/individual on the job must be able to:

- PC28.** avoid vibration in machining sequences.
- PC29.** apply burr-removal techniques on work pieces.
- PC30.** optimize machining strategies.
- PC31.** quickly react if anything goes wrong (making emergency stops)
- PC32.** obtain dimensions, geometries, surface roughness by interacting with PCB prototype machine.
- PC33.** make final parts conform to blueprints.
- PC34.** report health, safety, and environmental issues to appropriate personnel.
- PC35.** report equipment failure to appropriate personnel.

Complete the documentation

To be competent, the user/individual on the job must be able to:

- PC36.** note all production related information on respective register, log, work sheet.
- PC37.** update all production information on MIS system.
- PC38.** store Copper CAM program against batch number for future reference.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

Qualification Pack

- KU1.** PCB materials, types, and their applications in prototyping.
- KU2.** PCB prototype machine components and operational principles.
- KU3.** Reading and interpreting Gerber files, NC drill files, and CAD designs.
- KU4.** Setting up, calibrating, and configuring PCB prototype machines for machining.
- KU5.** drilling, milling, and engraving techniques for PCB fabrication.
- KU6.** how to optimize tool paths using CAM programming for precision and efficiency.
- KU7.** quality checks using metrology tools like calipers and optical inspection systems.
- KU8.** how to Identify and troubleshoot machining issues such as tool breakage or misalignment.
- KU9.** to conduct routine maintenance and ensuring machine safety protocols.
- KU10.** environmental and waste disposal guidelines for PCB manufacturing.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret Gerber files, NC drill files, and CAD designs for PCB prototyping.
- GS2.** setting up, calibrating, and operating PCB prototype machines (CNC, milling, and drilling).
- GS3.** optimize tool paths and programming for efficient PCB machining.
- GS4.** handle drilling, engraving, and cutting techniques for PCB fabrication.
- GS5.** use metrology tools (micrometers, calipers, optical inspection) for quality verification.
- GS6.** troubleshoot issues like tool misalignment, breakage, and machining errors.
- GS7.** do preventive maintenance procedures to ensure machine longevity and performance.
- GS8.** follow workplace safety protocols, including ESD (Electrostatic Discharge) protection.
- GS9.** keep technical documentation and report for machine settings, inspection results, and process deviations.
- GS10.** Collaborate and teamwork with design engineers, production, and quality control teams to improve efficiency.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Operate PCB prototype machining</i>	24	29	-	27
PC1. follow selected process strategy.	1	1	-	1
PC2. • follow a given process strategy (external CAM programme) • opening Gerber file containing one circuit layer • opening next layers (maximum 4), if any • detecting or plotting card cut-out contour • opening Excellon file for drillings, if any • aligning layers (automatic or manual) • plotting tracks that are centerline texts • calculating isolation contours	1	2	-	1
PC3. upload generated CAM programme to PCB prototype machine and perform test runs.	1	1	-	1
PC4. load G-Code for Top Layer PCB prototype machine and print the circuit.	1	1	-	1
PC5. stick two-layer FR4 Material (Copper board) on Machine bed using 2way tape or any other clamp.	1	2	-	1
PC6. mount and align selected bare PCB board.	1	1	-	1
PC7. plug engraving bit in spindle motor.	1	1	-	1
PC8. connect one terminal of leveler probe in engraving bit and another one terminal of leveler probe stick on FR4 material.	1	1	-	1
PC9. using the Hand-Remote adjusts the X, Y and Z axis and set origin point.	1	1	-	1
PC10. select X-axis and press “=0” button and set X = 0, Select Y-axis and press “=0” button and set Y = 0, select Z-axis and rotate hand-wheel in Hand-Remote and move to 2mm.Now the Z = 2	-	1	-	1
PC11. press the mode button and select PC mode.	-	1	-	1
PC12. press the “cycle start” button in PCB prototype machine and start the auto surface detection operation.	-	1	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. completing the surface detection, it will come to starting position with Z = 20mm. Then remove the probe and press “cycle start” button and it will start Engraving. After completing the Top Layer it will come to starting position.	1	1	-	1
PC14. change the drill bit and drill four corner of your circuit board with bed.	1	1	-	1
PC15. remove the copper board from bed and flip the copper. Fit the copper board on the bed with correct position.	1	1	-	1
PC16. set the origin position. Select the mode button to Hand Remote mode. Select “X” move to 80mm and press “=0” button	1	1	-	1
PC17. load G-Code for Bottom Layer and print your circuit.	1	1	-	1
PC18. plug engraving bit in spindle motor. And connect one terminal of leveler probe in engraving bit and another one terminal of leveler probe stick on FR4 material.	1	1	-	1
PC19. press “Auto detect” button in hand-Remote and the Z axis move and touch on surface of the copper glad and came up 10mm top from the surface.	1	1	-	1
PC20. select Z-axis and rotate hand-wheel in Hand-Remote and move to 2mm. Now the Z = 2	1	1	-	1
PC21. press mode button and select PC mode.	1	1	-	1
PC22. press the “cycle start” button in Mach3mill and start the auto surface detection operation.	1	1	-	1
PC23. after completing the surface detection, it will come to starting position with Z = 20mm. Then remove probe and press “cycle start” button and it will start Engraving.	1	1	-	1
PC24. after completing the Bottom Layer it will come to starting position.	1	1	-	1
PC25. load G-Code for Drilling & repeat the above process from PC 18 to 24.	1	1	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC26. load G-Code for Cutting & repeat the above process from PC 18 to 24	1	1	-	1
PC27. remove the PCB board and separate machined PCB circuit board.	1	1	-	1
<i>Complete the machining process</i>	4	8	-	3
PC28. avoid vibration in machining sequences.	1	1	-	1
PC29. apply burr-removal techniques on work pieces.	1	1	-	1
PC30. optimize machining strategies.	1	1	-	1
PC31. quickly react if anything goes wrong (making emergency stops)	-	1	-	-
PC32. obtain dimensions, geometries, surface roughness by interacting with PCB prototype machine.	-	1	-	-
PC33. make final parts conform to blueprints.	-	1	-	-
PC34. report health, safety, and environmental issues to appropriate personnel.	-	1	-	-
PC35. report equipment failure to appropriate personnel.	1	1	-	-
<i>Complete the documentation</i>	2	3	-	-
PC36. note all production related information on respective register, log, work sheet.	-	1	-	-
PC37. update all production information on MIS system.	1	1	-	-
PC38. store Copper CAM program against batch number for future reference.	1	1	-	-
NOS Total	30	40	-	30



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/ELE/N0103
NOS Name	Operate PCB prototype machine and complete the machining process
Sector	Electronics
Sub-Sector	
Occupation	PCB Fabrication and Verification
NSQF Level	5
Credits	4
Version	1.0
Last Reviewed Date	01/11/2023
Next Review Date	31/10/2026
NSQF Clearance Date	01/11/2023

Qualification Pack

CUTM/ELE/N0104: Inspect, Finalize, and deliver the parts.

Description

Fabrication and Verification Specialist focuses on performing quality inspections, identifying defects, ensuring compliance with design specifications, and documenting findings. It includes finalizing components, proper packaging, and coordinating logistics for safe and timely delivery while adhering to industry standards.

Scope

The scope covers the following :

- Fabrication and Verification Specialist covers performing quality inspections, verifying specifications, identifying defects, and ensuring compliance with industry standards. It includes finalizing PCB components, proper documentation, safe packaging, and coordinating logistics for accurate and timely delivery of finished parts.

Elements and Performance Criteria

Inspect machined products

To be competent, the user/individual on the job must be able to:

- PC1.** complete post processing methods.
- PC2.** clean parts for inspection.
- PC3.** check visual inspection on completion of process as per drawing.
- PC4.** check calibration status of various measuring instrument.
- PC5.** identify parameter wise measuring instruments for measurement.
- PC6.** inspect machined product with various measurement tools like Vernier caliper, micrometer.
- PC7.** check surface finish as per requirements.
- PC8.** check fitness of product using go or no-go devices.
- PC9.** check visual inspection on damage, burr, excess materials.
- PC10.** document measured data for various parameters.

Finalize and deliver the product

To be competent, the user/individual on the job must be able to:

- PC11.** clean parts.
- PC12.** pack parts in proper way to avoid any damages during transport.
- PC13.** complete delivery documents.
- PC14.** deliver parts, drawings, and digital memory devices to appropriate location and/or personnel as required by the organization.
- PC15.** dismount tools, clamping, accessories.
- PC16.** clean machines.
- PC17.** set environment to its initial state, ready for next job.
- PC18.** evaluate and report on factors and outcomes relevant to requirements and expectations.

Qualification Pack

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** inspection standards, tolerances, and quality benchmarks.
- KU2.** measurement tools like calipers, micrometers, and CMM (Coordinate Measuring Machines).
- KU3.** how to Interpret technical drawings, blueprints, and specifications for final inspection.
- KU4.** Identifying defects, deviations, and implementing corrective actions.
- KU5.** surface finish, dimensional accuracy, and assembly requirements.
- KU6.** Documentation of inspection reports and quality assurance records.
- KU7.** Packaging and handling techniques to prevent damage during transit.
- KU8.** Coordinating with logistics for timely and safe delivery of parts.
- KU9.** Ensuring workplace safety and adherence to environmental regulations.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret technical drawings, Gerber files, and PCB specifications.
- GS2.** use inspection tools like micrometers, calipers, coordinate measuring machines.
- GS3.** identify defects, deviations, and non-conformities in PCB components and assemblies.
- GS4.** document and report of inspection findings, rework recommendations, and quality reports.
- GS5.** manage time and workflow coordination to meet delivery schedules.
- GS6.** do proper packaging, labeling, and handling procedures to prevent damage during transportation.
- GS7.** follow safety protocols, ESD (Electrostatic Discharge) protection, and environmental compliance guidelines.
- GS8.** coordinate with design, production, and logistics teams to ensure timely part delivery.
- GS9.** address quality issues and suggest process improvements for finalizing PCB parts.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Inspect machined products</i>	22	14	-	20
PC1. complete post processing methods.	2	1	-	2
PC2. clean parts for inspection.	2	1	-	2
PC3. check visual inspection on completion of process as per drawing.	2	1	-	1
PC4. check calibration status of various measuring instrument.	3	1	-	2
PC5. identify parameter wise measuring instruments for measurement.	3	1	-	4
PC6. inspect machined product with various measurement tools like Vernier caliper, micrometer.	2	1	-	3
PC7. check surface finish as per requirements.	2	2	-	2
PC8. check fitness of product using go or no-go devices.	2	2	-	1
PC9. check visual inspection on damage, burr, excess materials.	2	2	-	2
PC10. document measured data for various parameters.	2	2	-	1
<i>Finalize and deliver the product</i>	18	16	-	10
PC11. clean parts.	2	2	-	2
PC12. pack parts in proper way to avoid any damages during transport.	2	2	-	1
PC13. complete delivery documents.	3	2	-	2
PC14. deliver parts, drawings, and digital memory devices to appropriate location and/or personnel as required by the organization.	3	3	-	1
PC15. dismount tools, clamping, accessories.	2	2	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC16. clean machines.	2	2	-	1
PC17. set environment to its initial state, ready for next job.	2	2	-	1
PC18. evaluate and report on factors and outcomes relevant to requirements and expectations.	2	1	-	1
NOS Total	40	30	-	30

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/ELE/N0104
NOS Name	Inspect, Finalize, and deliver the parts.
Sector	Electronics
Sub-Sector	
Occupation	PCB Fabrication and Verification
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	01/11/2023
Next Review Date	31/10/2026
NSQC Clearance Date	01/11/2023

Qualification Pack

CUTM/ELE/N0105: Implement organizational health, safety and security for self and others

Description

This NOS outlines the competencies required to ensure and implement workplace safety standards in PCB NPI (New Product Introduction) fabrication and verification. It includes compliance with safety regulations, independent handling of 3D printing and prototyping processes, and effective communication, while maintaining professionalism and continuous skill enhancement.

Scope

The scope covers the following :

- The scope of this NOS covers safety management and best practices in PCB prototyping using 3D printers and related tools. It includes parameter setting, tool handling, interpreting instructions, maintaining workplace hygiene, and engaging clients professionally. It supports efficient, secure, and quality-driven PCB new product development workflows.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** effectively apply health and safety regulations in workplace
- PC2.** proactively promote best practice in health and safety in working environment
- PC3.** work independently on 3D printer machine
- PC4.** create manual and programs for various types of prototypes
- PC5.** select suitable machine parameters
- PC6.** select and set most appropriate tools for planned work
- PC7.** maintain tools to ensure they are in best condition
- PC8.** communicate and collaborate effectively with colleagues, team members, and other professionals
- PC9.** communicate effectively, using strong inter-personal Skills with co-workers, clients, and other related professionals to ensure that project meets the requirement
- PC10.** interpret and extract technical data and instructions from any available sources
- PC11.** employ discretion and confidentiality when dealing with clients
- PC12.** clarify terms of reference, specifications, and instructions, for accurate implementation of requirements
- PC13.** maintain proactive continuous professional development in order to sustain knowledge and skill in new and developing technologies and practices
- PC14.** engage with customers effectively by prioritizing their needs
- PC15.** analyse manufacturing feasibility
- PC16.** demonstrate high levels of critical thinking

Knowledge and Understanding (KU)

Qualification Pack

The individual on the job needs to know and understand:

- KU1.** Importance of adhering to workplace health and safety regulations and their impact on operations.
- KU2.** Best practices in ensuring a hazard-free environment in PCB fabrication and prototyping setups.
- KU3.** Working principles, operation, and maintenance of 3D printers and their role in PCB prototyping.
- KU4.** Types of programs and software used to create manual and automated prototypes.
- KU5.** Machine parameter selection based on design specifications and materials.
- KU6.** Tool types and their suitability for different PCB fabrication tasks.
- KU7.** Standard procedures for maintaining tools and ensuring optimal condition.
- KU8.** Techniques for professional communication and collaboration with team members and clients.
- KU9.** Methods to interpret technical instructions, drawings, and terms of reference.
- KU10.** Importance of professional development to stay updated with technological advancements in PCB manufacturing.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Apply health and safety measures efficiently in high-tech manufacturing environments.
- GS2.** Promote a proactive safety culture through leadership and example.
- GS3.** Operate and troubleshoot 3D printing equipment for PCB prototype development independently.
- GS4.** Create, run, and modify manual and digital programs for prototyping and testing.
- GS5.** Select appropriate tools and machine settings based on job requirements.
- GS6.** Maintain and service equipment regularly to avoid failures and defects.
- GS7.** Collaborate with technical teams and clients to ensure alignment with project goals.
- GS8.** Exhibit discretion and professionalism in handling sensitive technical data.
- GS9.** Analyze technical feasibility and clarify instructions for efficient execution.
- GS10.** Demonstrate critical thinking and decision-making in resolving manufacturing challenges.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	40	-	30
PC1. effectively apply health and safety regulations in workplace	2	2	-	3
PC2. proactively promote best practice in health and safety in working environment	2	2	-	4
PC3. work independently on 3D printer machine	2	2	-	1
PC4. create manual and programs for various types of prototypes	2	2	-	1
PC5. select suitable machine parameters	2	2	-	1
PC6. select and set most appropriate tools for planned work	2	2	-	1
PC7. maintain tools to ensure they are in best condition	2	2	-	1
PC8. communicate and collaborate effectively with colleagues, team members, and other professionals	2	2	-	2
PC9. communicate effectively, using strong inter-personal Skills with co-workers, clients, and other related professionals to ensure that project meets the requirement	2	2	-	2
PC10. interpret and extract technical data and instructions from any available sources	2	2	-	2
PC11. employ discretion and confidentiality when dealing with clients	2	2	-	2
PC12. clarify terms of reference, specifications, and instructions, for accurate implementation of requirements	2	3	-	2
PC13. maintain proactive continuous professional development in order to sustain knowledge and skill in new and developing technologies and practices	2	4	-	2

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. engage with customers effectively by prioritizing their needs	2	6	-	2
PC15. analyse manufacturing feasibility	1	3	-	2
PC16. demonstrate high levels of critical thinking	1	2	-	2
NOS Total	30	40	-	30



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/ELE/N0105
NOS Name	Implement organizational health, safety and security for self and others
Sector	Electronics
Sub-Sector	
Occupation	PCB Fabrication and Verification
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	01/11/2023
Next Review Date	31/10/2026
NSQF Clearance Date	01/11/2023



Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Assessment System Overview:

- Batches assigned to the assessment agencies for conducting the assessment on SIDH or email
- Assessment agencies send the assessment confirmation to TP looping AB/AA to accept in SIDH
- Assessment agency deploys the ToA certified Assessor for executing the assessment
- AB monitors the assessment process & records

2. Testing Environment:

- Check the Assessment location, date and time
- If the batch size is more than 30, then there should be 2 Assessors.



Qualification Pack

- Check that the allotted time to the candidates to complete Theory & Practical Assessment is correct.

3. Assessment Quality Assurance levels/Framework:

- Question bank is created by the Subject Matter Experts (SME) are verified by the AB
- Questions are mapped to the specified assessment criteria.
- Assessor must be ToA certified & trainer must be ToT Certified

4. Types of evidence or evidence-gathering protocol:

- Time-stamped & geotagged reporting of the assessor from assessment location.
- Centre photographs with signboards and scheme specific branding
- Video Call verification/online proctoring will be done during the assessment from AB.

5. Method of verification or validation:

- Surprise visit to the assessment location

6. Minimum Percentage Required

- 70% overall to pass.
- Minimum 70% in each NOS

7. Reassessment: Candidates failing the assessment will be given one opportunity for reassessment after additional training.

8. Assessor Details: ToA Certified Assessor with relevant experience

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Qualification Pack

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CUTM/ELE/N0101.Interpretation of designs and drawings and process planning	30	40	-	30	100	15
CUTM/ELE/N0102.Copper CAM Programming, Metrology, Maintenance, and repair of machine.	40	60	-	-	100	20
CUTM/ELE/N0103.Operate PCB prototype machine and complete the machining process	30	40	-	30	100	20
CUTM/ELE/N0104.Inspect, Finalize, and deliver the parts.	40	30	-	30	100	20
CUTM/ELE/N0105.Implement organizational health, safety and security for self and others	30	40	-	30	100	15
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	190	240	-	120	550	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.